

Point-process TDA: status

September 16, 2026

1 Summary

- Classical summary functions beat persistence images on every parameter-estimation family and on classification overall (accuracy 0.719 vs. 0.666).
- **The one place PH beats a classical model:** classification in the two most structured bands, against L alone (the published baseline): +1.8 and +3.7 points of accuracy. It is behind the best classical model in every band.
- On parameter estimation PH is behind L alone in every band on every family.
- The PH deficit is concentrated near Poisson, where the classical 5% test itself cannot detect a departure: -11.6 points against L alone.
- H_0 carries most of the persistence information; H_1 adds little, and on two families slightly hurts.
- **The PH numbers are a lower bound.** Every PH run so far uses one filtration scale (DTM, $k = 5$), and for classification omits the point count $n(x)$ that every classical model receives.

2 Data and methods

- **Models:** Poisson, Thomas, nested Thomas, Matérn II (hard-core), LGCP on the unit square, with expected count $\bar{n} \in [100, 800]$.
- **Distance from Poisson, $\tilde{\delta}$:** the largest standardised deviation of the model's theoretical L -function from Poisson, divided by the 5% critical value of the classical global envelope test. So $\tilde{\delta} \leq 1$ means exactly that the classical test does not reject Poisson.
- **Training:** 10,000 patterns per model from the prior (fixed 85/15 train/validation split).
- **Test set:** 50,000 patterns, 2,500 in every cell of 5 models $\times$ 4 bands: **below** ($\tilde{\delta} < 1$), **weak** (1–2), **moderate** (2–4), **strong** (≥ 4). Poisson patterns are split evenly across the bands so chance is 0.20 in each. Selected from patterns that already existed; no model was retrained. Details: `docs/test_set.md`.
- **Classical:** 1-D CNN of Vihrs (2022) on summary functions plus $n(x)$. All 15 subsets of $\{L, F, G, J\}$ were run.
- **PH:** CNN on persistence images of the DTM filtration ($k = 5$): H_0 , H_1 , or both as two channels; plus $n(x)$ for parameter estimation only.
- **Training:** early stopping on validation loss; mean $\pm$ SD over 10 seeds.

3 Parameter estimation

Metric: MSE of the standardised log-parameters, averaged over parameters. Lower is better; ≈ 1 means predicting the training mean. Not comparable across families.

Model	Thomas	Nested	Matérn II	LGCP
L	0.1690 ± 0.0082	0.2392 ± 0.0081	0.0140 ± 0.0005	0.1720 ± 0.0024
G	0.4735 ± 0.0191	0.2528 ± 0.0114	0.0135 ± 0.0002	0.2303 ± 0.0023
$L+F+G$	0.1677 ± 0.0090	0.2287 ± 0.0130	0.0140 ± 0.0003	0.1724 ± 0.0039
$L+F+G+J$	0.1697 ± 0.0076	0.2333 ± 0.0108	0.0137 ± 0.0005	0.1706 ± 0.0037
PH H_0	0.2487 ± 0.0082	0.2867 ± 0.0125	0.0361 ± 0.0017	0.1877 ± 0.0056
PH H_1	0.3701 ± 0.0199	0.4143 ± 0.0172	0.2692 ± 0.0243	0.2264 ± 0.0043
PH H_0+H_1	0.2308 ± 0.0167	0.3022 ± 0.0126	0.0373 ± 0.0022	0.1834 ± 0.0038

Table 1: Parameter estimation, normalised loss on the test set (mean $\pm$ SD over 10 seeds). Shown: L alone and every subset of $\{L, F, G, J\}$ that is best on at least one family. Bold: best per family.

Family	Model	Distance from Poisson			
		below	weak	moderate	strong
Thomas	L	0.282	0.234	0.115	0.045
	$L+F+G$ (best)	0.284	0.236	0.109	0.042
	PH H_0	0.328	0.357	0.228	0.081
	PH H_1	0.444	0.451	0.381	0.204
	PH H_0+H_1	0.320	0.329	0.202	0.072
Nested	L	0.340	0.285	0.185	0.146
	$L+F+G$ (best)	0.331	0.274	0.175	0.135
	PH H_0	0.371	0.361	0.238	0.177
	PH H_1	0.429	0.452	0.401	0.375
	PH H_0+H_1	0.392	0.379	0.250	0.188
Matérn II	L	0.028	0.013	0.008	0.007
	G (best)	0.028	0.013	0.007	0.006
	PH H_0	0.058	0.049	0.027	0.011
	PH H_1	0.234	0.078	0.338	0.427
	PH H_0+H_1	0.061	0.050	0.027	0.012
LGCP	L	0.218	0.177	0.156	0.137
	$L+F+G+J$ (best)	0.217	0.174	0.156	0.135
	PH H_0	0.230	0.191	0.173	0.158
	PH H_1	0.254	0.213	0.220	0.219
	PH H_0+H_1	0.222	0.192	0.171	0.149

Table 2: Parameter estimation by band (normalised loss, mean over 10 seeds; 2,500 patterns per band per family).

Family		Distance from Poisson			
		below	weak	moderate	strong
Thomas	vs. best	1.12	1.40	1.85	1.73
	vs. L	1.13	1.41	1.75	1.60
Nested	vs. best	1.18	1.38	1.42	1.40
	vs. L	1.15	1.33	1.35	1.28
Matérn II	vs. best	2.17	3.97	3.66	1.93
	vs. L	2.16	3.90	3.35	1.69
LGCP	vs. best	1.02	1.10	1.10	1.10
	vs. L	1.02	1.08	1.10	1.09

Table 3: Gap between PH and the classical models: loss of PH H_0+H_1 divided by the loss of the family’s best classical model, and of L alone (1 = parity, > 1 = PH worse). Bold: largest gap against the best model, per family.

- The best classical subset differs by family ($L+F+G$ on Thomas and nested, G on Matérn II, $L+F+G+J$ on LGCP), but is within about one seed SD of L alone except on nested (0.229 vs. 0.239).
- PH (H_0+H_1) has $1.1\times$ (LGCP) to $2.8\times$ (Matérn II) the loss of the best classical model.
- The gap (Table 3) is smallest near Poisson (1.02–1.18) except on Matérn II (2.17), peaks at **moderate** on Thomas and nested and at **weak** on Matérn II, and is flat at ≈ 1.1 on LGCP.
- Against L alone the ratios are similar, and above 1 in every cell: PH never beats the published baseline on parameter estimation.
- Matérn II is the largest gap (up to $4\times$). DTM with $k = 5$ averages over five neighbours and so smooths away the minimum interpoint distance that defines a hard core.
- H_1 alone is weak. Adding it to H_0 helps on Thomas (0.231 vs. 0.249) and LGCP, and slightly hurts on nested (0.302 vs. 0.287) and Matérn II.

4 Classification

5-way classification (Poisson, Thomas, nested, Matérn II, LGCP), all models trained on the same pooled training set. Metric: accuracy (chance 0.20 in every band).

Model	Accuracy	Recall per family				
		Poisson	Thomas	Nested	Matérn II	LGCP
L	0.685 ± 0.004	0.807	0.421	0.556	0.949	0.691
$L+J$	0.717 ± 0.003	0.824	0.475	0.603	0.941	0.741
$L+F+G$	0.719 ± 0.003	0.813	0.478	0.613	0.945	0.747
$L+F+J$	0.717 ± 0.006	0.811	0.474	0.611	0.943	0.747
PH H_0	0.661 ± 0.004	0.825	0.371	0.572	0.807	0.729
PH H_1	0.414 ± 0.004	0.668	0.190	0.294	0.382	0.537
PH H_0+H_1	0.666 ± 0.003	0.832	0.389	0.586	0.799	0.725

Table 4: Classification on the test set (mean over 10 seeds; SD shown for accuracy). Shown: L alone and the three most accurate subsets of $\{L, F, G, J\}$. Bold: best per column among the models shown.

Model	Accuracy				Detection power			
	below	weak	mod.	strong	below	weak	mod.	strong
L	0.474	0.659	0.766	0.841	0.439	0.908	0.993	0.999
$L+J$	0.471	0.684	0.815	0.898	0.427	0.904	0.993	1.000
$L+F+G$	0.473	0.684	0.819	0.899	0.432	0.909	0.994	1.000
$L+F+J$	0.471	0.682	0.820	0.895	0.426	0.901	0.994	1.000
PH H_0	0.356	0.639	0.776	0.872	0.299	0.809	0.978	0.999
PH H_1	0.257	0.332	0.434	0.634	0.168	0.433	0.650	0.804
PH H_0+H_1	0.358	0.643	0.784	0.878	0.313	0.821	0.978	1.000

Table 5: Left: accuracy by band (chance 0.20 in every band). Right: the classifiers used as tests of “Poisson vs. not Poisson” (score $1 - P(\text{Poisson})$), threshold set at 5% false alarms per $\bar{n}$ octave on half of the test set’s Poisson patterns, power measured on the structured patterns; the false-alarm rate on the other half is 0.045–0.049 for every model.

	below	weak	moderate	strong
<i>vs. the best classical model ($L+F+G$)</i>				
Accuracy	−11.5	−4.1	−3.5	−2.1
Detection power	−11.9	−8.8	−1.6	0.0
<i>vs. L alone (the published baseline)</i>				
Accuracy	−11.6	−1.5	+1.8	+3.7
Detection power	−12.6	−8.7	−1.5	0.0

Table 6: Gap between PH H_0+H_1 and the classical models, in percentage points (negative = PH worse, positive = PH better). Bold: largest deficit against the best model, per row.

- $L+F+G$ is the best classifier (0.719), with $L+J$ and $L+F+J$ at 0.717. L alone: 0.685.
- PH H_0+H_1 reaches 0.666: 5.3 points behind the best model and 1.9 behind L alone. The deficit is on Matérn II (recall 0.799 vs. 0.945) and Thomas (0.389 vs. 0.478). PH has the best Poisson recall of any model shown.
- **By band, against L alone:** $-11.6 / -1.5 / +1.8 / +3.7$ points. PH overtakes the published baseline once the structure is comfortably detectable. Against the best classical model it stays behind in every band, by 2.1 points at **strong**.
- Near Poisson PH is conservative: it labels 52% of the structured **below** patterns as Poisson, against 35% for $L+F+G$. That is both why its Poisson recall is highest and why its **below** accuracy is lowest.
- As a Poisson test every model holds the 5% false-alarm rate. PH has about 9–13 points less power in **below** and **weak**; from **moderate** every model except H_1 detects essentially everything.
- H_1 alone is above chance (0.414) but the weakest classifier and detector. Adding it to H_0 is worth half a point (0.666 vs. 0.661).

5 Next

- Classification PH with $n(x)$, H_0 and H_0+H_1 (ready to submit: `slurm/dv3_pi_logn_submit.sh`). Removes the one asymmetry in Table 4.
- Multi-scale DTM ($k = 5, 10, 15$); needs the $k = 10, 15$ diagrams.
- An alpha or Čech filtration channel for Matérn II.